Earners AI and ML by Aditya Jain sir - Lecture 12

Summary Notes

Link to Telegram Community: https://t.me/EarnersAdityaJain

```
for elem in s:
  print(elem)
```

```
A
d
i
t
y
a

J
a
i
n
```

```
Start coding or generate with AI.
```

Recap - Strings

```
s = "Aditya Jain"
type(s)
```

```
str
```

```
n = len(s)
print(n)
```

```
11
```

```
s[2]
```

```
'i'
```

```
s[-1]
```

```
'n'
```

```
print(s)
```

```
Aditya Jain
```

```
a1 = s[2:7]
```

```
print(a1)
```

```
itya
```

```
print(s[:6])
```

```
Aditya
```

```
for elem in s:
  print(elem)
```

```
A
d
i
t
y
a
```

```
J
a
i
n
```

```
n = len(s)
for i in range(n):
  print(s[i], end = " ")
```

```
A d i t y a   J a i n
```

```
print(s)
```

```
Aditya Jain
```

```
a = s.split(" ")
print(a)
```

```
['Aditya', 'Jain']
```

```
s = "Aditya Jain likes to do programming"
a = s.split(" ")
print(a)
```

```
['Aditya', 'Jain', 'likes', 'to', 'do', 'programming']
```

```
for elem in a:
  print(elem)
```

```
Aditya
Jain
likes
to
do
programming
```

Double-click (or enter) to edit

HW - DPP- Tricky Advanced Questions

```
text = "Programming"

count = 0

for i in range(len(text)):
    if text[i] == text[-1]:
        count += 1

print(count)
```

```
2
```

```
Start coding or generate with AI.
```

```
word = "PYTHON"

result = ""

for i in range(len(word)):
    if i % 2 == 1:
        result += word[i]

print(result)
```

```
YHN
```

```
Start coding or generate with AI.
```

```
text = "Data"
```

```
result = ""

for ch in text:
    result = ch + result

print(result)
```

```
ataD
```

```
Start coding or generate with AI.
```

```
text = "Data"

result = ""

for ch in text:
    result = result + ch

print(result)
```

```
Data
```

```
Start coding or generate with AI.
```

```
word = "Machine"

count = 0

for ch in word:
    if ch not in "aeiAEouIOU":
        count += 1

print(count)
```

```
4
```

```
Start coding or generate with AI.
```

```
word = "Python"

i = len(word) - 1

while i >= 0:
    print(word[i], end="")
    i -= 2
```

```
nhy
```

```
Start coding or generate with AI.
```

```
text = "AI"

result = ""

for i in range(4):
    result = result + text
print(result)
print(len(result))
```

```
AIAIAIAI
8
```

```
Start coding or generate with AI.
```

```
word = "Computer"

count = 0

for ch in word:
    if ch < "m":
      print(ch)
      count += 1
```

```
print(count)
```

```
C
e
2
```

```
ord('A')
```

```
65
```

```
ord('B')
```

```
66
```

```
ord('Z')
```

```
90
```

```
ord('a')
```

```
97
```

```
ord('z')
```

```
122
```

```
chr(122)
```

```
'z'
```

```
chr(97)
```

```
'a'
```

```
chr(65)
```

```
'A'
```

```
chr(128522)
```

```
'😊'
```

```
Start coding or generate with AI.
```

```
text = "Python"

result = ""

for i in range(len(text)):
    if text[i] in "aeiouAEIOU":
        result += text[i]

print(result)
```

```
o
```

```
Start coding or generate with AI.
```

```
word = "LEVEL"

same = 0

for i in range(len(word)):
    if word[i] == word[-(i + 1)]:
      print(i, -(i+1), "--->", word[i], word[-(i+1)])
      same += 1

print(same)
```

```
0 -1 ---> L L
1 -2 ---> E E
2 -3 ---> V V
3 -4 ---> E E
4 -5 ---> L L
5
```

```
Start coding or generate with AI.
```

```
text = "Python"

result = ""

for i in range(len(text)):
    if i < len(text) // 2:
        result += text[i]

print(result)
```

```
Pyt
```

Functions

```
print("Aditya")
```

```
Aditya
```

```
len("AJ")
```

```
2
```

```
type(2)
```

```
int
```

```
# def function_name():
#   #definition
```

```
def greet():
  print("Hello")
```

```
greet() #function call
```

```
Hello
```

```
#Write a function that prints "Hello name" for whatever name you want
```

```
greet()
```

```
Hello
```

```
greet()
```

```
Hello
```

```
#code reusability

# function definition ---> parameter
# function call --> argument
```

```
def greet(name):
  print("Hello", name)
```

```
greet("AJ")
```

```
Hello AJ
```

```
greet("Akshat")
```

```
Hello Akshat
```

```
greet("Nidhi")
```

```
Hello Nidhi
```

```
greet("Aditya")
```

```
Hello Aditya
```

Positional Arguments

```
def show(name, age):
  print("Name is", name)
  print("Age is", age)
```

```
show("Raj", 22)
```

```
Name is Raj
Age is 22
```

```
show("Priya", 20)
```

```
Name is Priya
Age is 20
```

```
show(20, "Raj")
```

```
Name is 20
Age is Raj
```

Keyworded arguments

```
def show(name, age):
  print("Name is", name)
  print("Age is", age)
```

```
show(name = "Raj" , age = 20)
```

```
Name is Raj
Age is 20
```

```
show(age = 20, name = "Raj")
```

```
Name is Raj
Age is 20
```

Return keyword

```
def show(a,b):
  return a+b
```

```
s = show(2,3)
print(s)
```

```
5
```

Quiz

```
def greet():
    print("Hello")
```

```
greet()
```

```
Hello
```

```
def greet():
  print("Welcome")

greet()
greet()
```

```
Welcome
Welcome
```

```
def add(a, b):
    print(a + b)

add(10, 20)
```

```
30
```

```
def square(n):
    return n * n

print(square(5))
```

```
25
```

```
def message():
    print("Python")

x = message()

print(x)
```

```
Python
None
```

```
def multiply(a, b):
    return a * b

result = multiply(4, 5)

print(result)
```

```
20
```

HW

```
def check(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"

print(check(15))
```

```
def greet(name):
    print("Hello", name)

greet("Rahul")
```

```
def total(numbers):
    s = 0

    for n in numbers:
        s += n

    return s

print(total([5, 10, 15]))
```

```
def display():
    return "Python"

print(display())
```

```
def largest(a, b):
    if a > b:
        return a
    else:
        return b

print(largest(12, 25))
```

```
def stars():
    print("*" * 5)

for i in range(3):
    stars()
```